MOVIZ – Scientific Model Visualization

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INTRODUCTION

- Computational models help explore complex system hypotheses, but sharing and maintaining model implementations is difficult due to scattered scientific knowledge.
- The SKEMA framework contributes to machine-assisted knowledge discovery, curation, and modeling, enabling experts to manage diverse data, knowledge, and models with traceability.
- The framework applies models and simulators to various use cases, with a focus on modeling COVID-19 and space weather's causes and impacts.

BACKGROUND

- The models extracted have multiple layers and syntactic foundations and relations.
- There is no one framework to represent all models.



- **GroMEt (Grounded Model Exchange):**
  - Multi-model representation
  - Accommodates function networks, ordinary differential equations, and agent-based models
  - Designed for multiple modeling frameworks
  - GroMEt function networks are exported as JSON
  - For larger models, these exports are of considerable size
  - The structure of these exports is not easily comprehensible for humans
  - A schema has been designed to visualize all components of the function networks

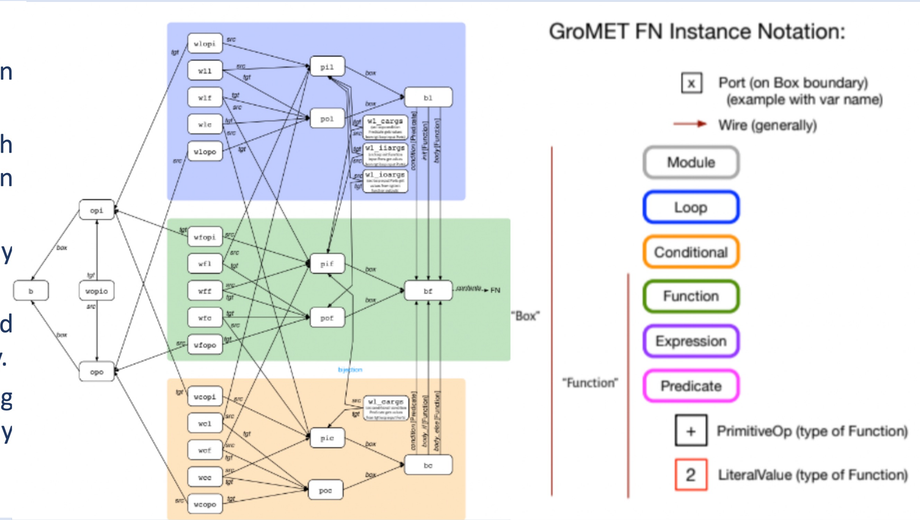


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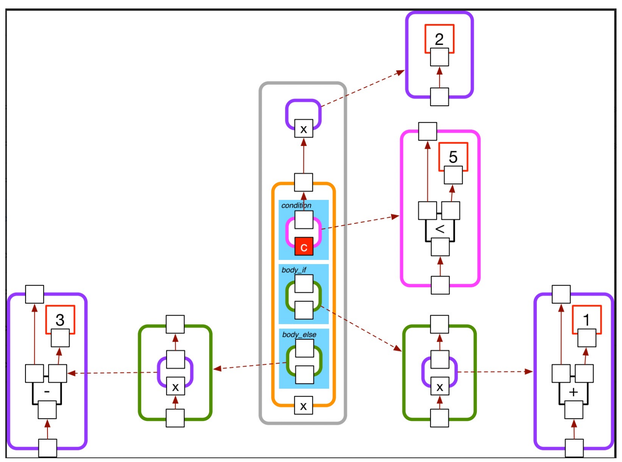
Method

MOVIZ (Model Visualization):

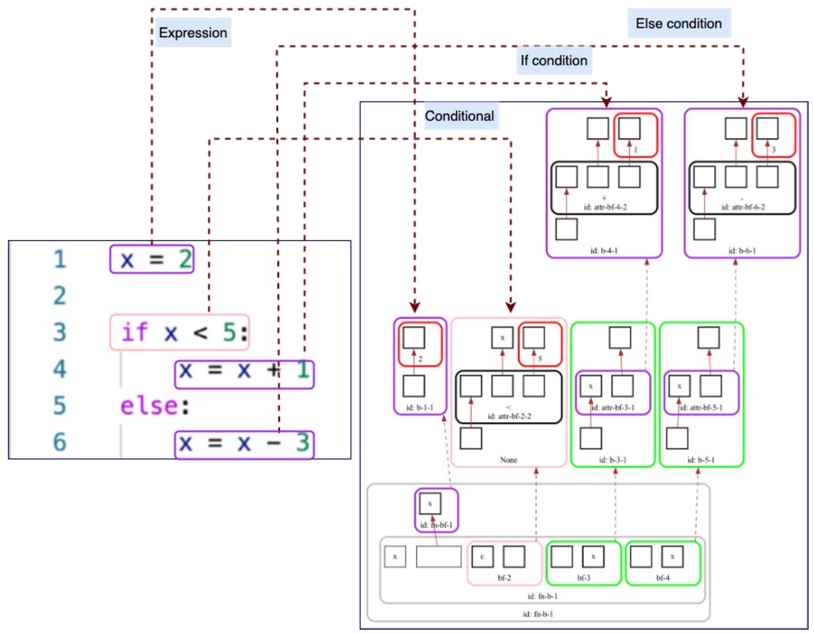
- MOVIZ tool visualizes computation graphs in scientific model codebases
- Uses scalable techniques to automate graph construction and manage visualization complexity
- Visualization reveals insights not easily accessible through raw source code
- Employs a hybrid approach of direct and declarative manipulation to manage complexity.
- Visualizations were previously drawn using OmniGraffle, but are now automatically rendered



Results



Manual diagram for a conditional using OmniGraffle



Automated diagram for a conditional using MOVIZ

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